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Team no: 10

Team Name: patches

Total Marks: 100

Due 31 March 2026

HCI & Computer Graphics

Milestone 1

Initial Concept and Needs Finding Plans

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Section: C

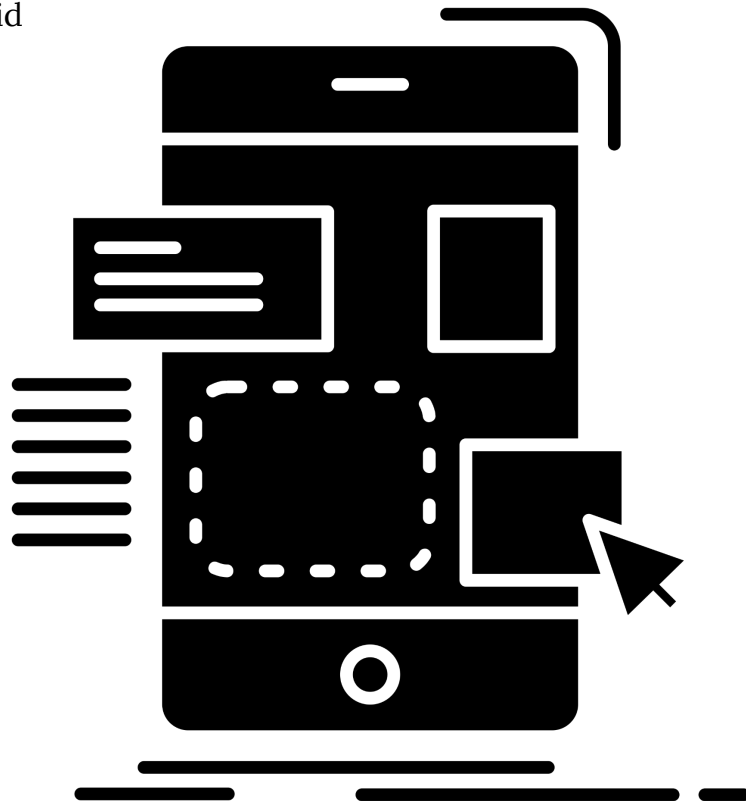


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Milestone 1: Initial Concept and Needs Finding Plans

HCI & Computer Graphics Course Project Documentation

Section 1: Design Brief

Target User Population

The target users encompass a broad demographic spectrum in Pakistan, divided into two primary archetypes. First: Homemakers and heads of households with varying levels of digital literacy who manage daily local expenses. Second: University students, freelancers, and small business owners (SMEs) who require granular financial oversight and rapid, frictionless logging of diverse receipts.

Problem to be Solved

The core activity to support is the daily, accurate logging of financial expenses. Currently, manual financial tracking imposes a high cognitive load, leading to inconsistent logging and high user drop-off. A critical usability barrier is that standard digital tools rely on rigid manual entry or basic OCR that fails to process the informal, handwritten, or faded physical receipts prevalent in Pakistani retail (e.g., local dairies, neighborhood hardware stores). This renders current finance apps intimidating and practically useless for non-expert users attempting to manage daily budgets.

Competitive Landscape

The current landscape consists of a mix of digital and analogue tools:

- **Analogue Methods:** Physical "Khata" notebooks, saving loose receipts in drawers, and mental math. While highly accessible, they lack searchability, aggregation, and analytics.
- **Digital Tools:** Standard banking apps (which only track digital payments), generic OCR scanners (which fail on handwritten Urdu/English mix), and

complex apps like Cashew or Wallet (which require tedious manual categorization).

None of these seamlessly bridge the gap between physical, messy local receipts and structured digital analytics without significant user effort.

System Goals

DailyKhata is an inclusive, multimodal AI expense management mobile application.

- **Capabilities:** It will replace manual entry with a frictionless camera-to-AI pipeline capable of parsing unstructured, handwritten local receipts. For tech-novices, it includes a "Voice-First" NLP interaction model to log expenses via natural conversational phrasing. For power users, it features a responsive "Live Spreadsheet" UI for high-density data management.
 - **Advantage:** It bridges the digital divide by adapting to the user's digital literacy level. By emphasizing rapid error recovery (touch-to-edit verification) and utilizing generative AI for extraction, it drastically reduces the time and friction required to maintain an accurate financial ledger compared to traditional apps or paper methods.
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Section 2: Needs Finding Study Plan

High-Level Study Goals

The primary goal is to understand how Pakistani homemakers and students currently track daily expenses across informal and formal retail channels. We aim to identify the specific friction points that cause them to abandon financial tracking, and observe their mental models regarding how they categorize, verify, and store physical receipts.

Recruiting Criteria and Strategy

We will recruit 4 participants using convenience and snowball sampling.

Criteria:

- **Group A (2 users):** Homemakers or local shop frequenters who primarily use analogue methods (notebooks) or struggle with complex digital tracking.
- **Group B (2 users):** University students or freelancers who handle personal budgets and have attempted (but perhaps abandoned) using a digital finance app.

All participants must actively manage a monthly budget and regularly receive physical receipts. Interviews will last approximately 30-45 minutes and include workflow observations.

Interview Protocol

Overarching Question

How do users across different digital literacy levels currently track, manage, and abandon their daily expense logs, particularly when dealing with physical or informal receipts?

Introduction

Hi, we're Saad and Khadijah, and we're here to understand better how you manage your daily expenses and keep track of where your money goes. We are designing an app to make financial tracking completely frictionless, and we'd love to learn from your daily routines. This interview will take about 10-20 minutes. To the extent possible, We will keep your comments confidential, and any quotes used in our project will be entirely anonymized. This interview is completely voluntary, if you want to stop or skip a question at any time, just let us know. Do you mind if we take an audio recording just so we don't miss any of your insights? Do you have any questions before we begin?

Warm-up

Tell us a bit about your typical week, who handles the primary shopping and expense tracking in your household?

- *Follow-up:* What tools do you currently use to keep track of this money?
- *Follow-up:* Are you generally satisfied with how you keep track of your budget right now?

- *Follow-up:* What is the most frustrating part of managing your money?

Recent Concrete Instance: The Shopping Trip

We'd like you to think back to the last time you went grocery shopping or bought supplies from a local market. Can you walk us through what happened after you paid?

- *Follow-up:* Did you receive a receipt? Was it printed or handwritten?
- *Follow-up:* Where exactly did you put that receipt when you got home?
- *Follow-up:* Did you write the expense down somewhere? When and how?

Workflow Observation

We'd like to see how you figure out what you've spent this month. Can you show us the notebook, app, or system you use and talk us through your process right now?

- *Follow-up: (Observation)* If you had a receipt for "milk and bread" from a local dairy right now, show us exactly how you would record it in your system.
- *Follow-up:* How do you categorize things? Do you group them, or just write a total list?
- *Follow-up:* How much time would you estimate you spend doing this every week?

Friction and Abandonment

Think about a time recently when you *wanted* to record an expense but didn't. What stopped you?

- *Follow-up:* Was it because you lost the receipt, or was it too much effort?
- *Follow-up:* How do you handle situations where you don't get a receipt at all, like paying a rickshaw?
- *Follow-up:* Have you ever tried an expense-tracking mobile app? If you stopped using it, what made you quit?

Concept Introduction (Voice & AI)

If you had a tool where you could just take a photo of a messy receipt, or simply speak to your phone and say "I spent 500 on vegetables," how would that change your current routine?

- *Follow-up:* Do you think you would trust an AI to read your receipt correctly?
- *Follow-up:* How would you want the app to behave if it wasn't sure about a handwritten number?
- *Follow-up:* Would you prefer typing, speaking, or snapping photos for your daily logging?

Conclusion

Thank you so much for your time and for sharing your workflow with me. Your insights will be incredibly helpful for our project. Is there anything else about managing expenses or using finance apps that we haven't covered today? If you think of anything else, feel free to reach out.